

SKILL – 4: Spring Dependency Injection – Constructor & Setter Injection

#SKILL - 4 → Spring Dependency Injection – Constructor & Setter Injection

Prerequisites:

- Basic knowledge of Java
- General Idea on Spring Framework
- Basic Idea on DI & IoC

A training institute wants to automate the management of student information in a Spring application. The system should inject details such as **studentId**, **name**, **course**, and **academic year** into objects without manually assigning values. This Skill demonstrates **Constructor Injection** and **Setter Injection** using both **XML** and **Annotation** configurations.

Tasks:

1. Create a Java class named **Student** with fields: **studentId**, **name**, **course**, **year**.
2. Create a **Constructor** that accepts all fields and assigns values.
3. Create **Setter methods** for at least two fields (example: **course**, **year**).
4. Configure the Student bean using:
 - a. **XML Configuration** - The student should take:
 - a **POJO class**
 - an **XML configuration file**
 - a **main class** to load the container
 - b. **Annotation Configuration** - The student should take:
 - a **POJO class**
 - a **Java class with necessary annotations**
 - a **main class** to load the container
5. Create a Spring configuration file (XML or Java-based).
6. Write a main class to load the Spring IoC container.
7. Retrieve the Student bean and print the injected values.
8. **Push the Spring (core) project to GitHub.**

You can use both:

- applicationContext.xml (XML autowiring)
- Annotations (@Autowired, @Component, etc.)

Spring supports mixed configuration.

Concept	Meaning	Spring Annotation
IoC	Spring controls object creation	(Internal)
DI	Injection of dependencies	@Autowired
Autowiring	Automatic DI process	@Autowired
Bean	Object managed by Spring	@Component, @Service

Create Project In Eclipse / STS

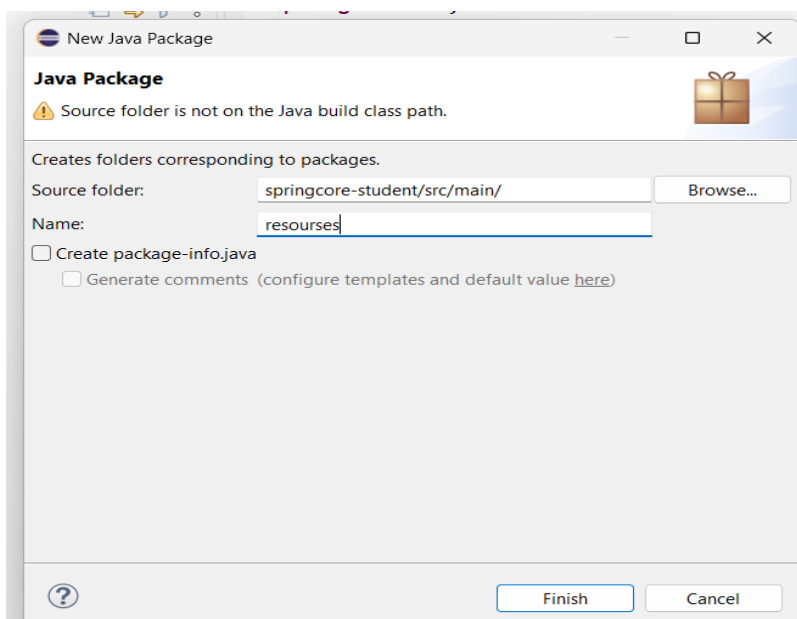
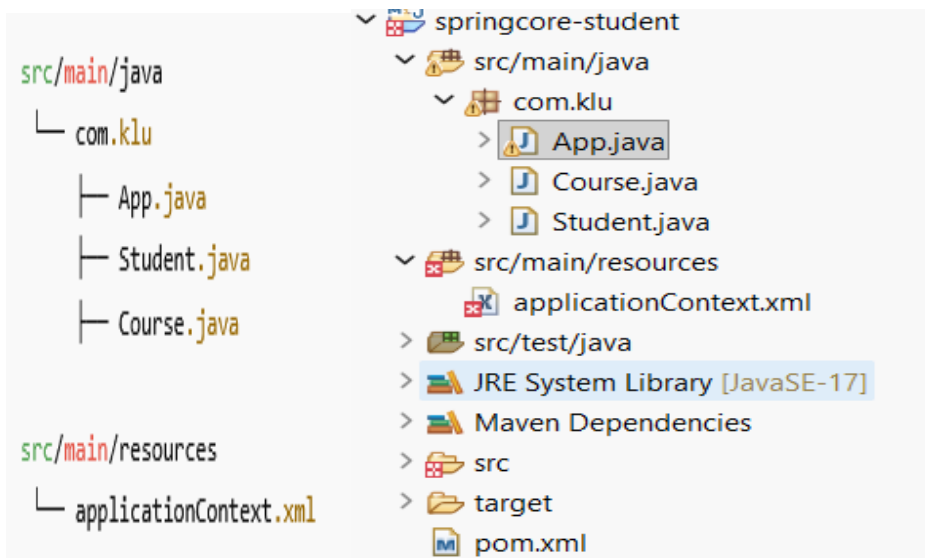
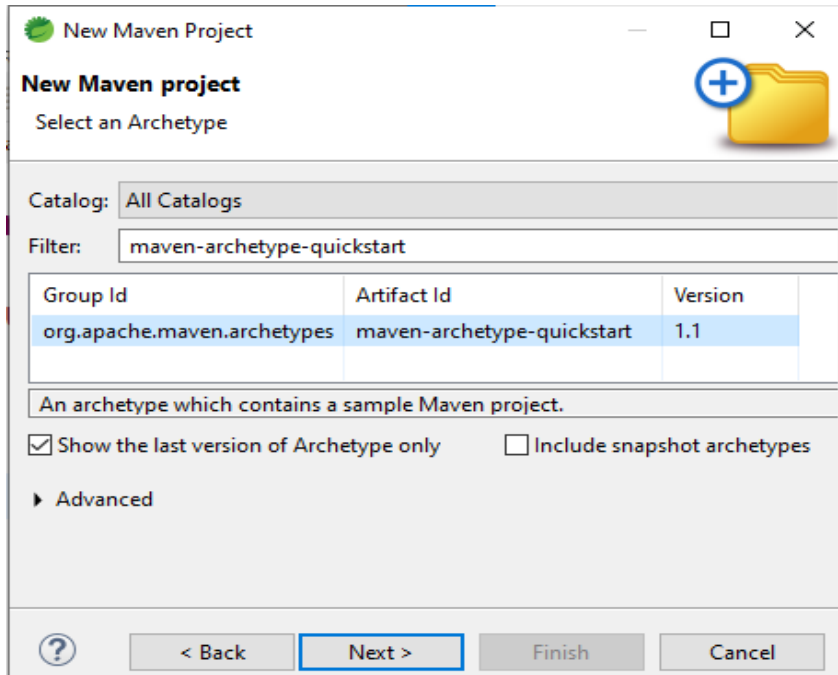
File → New → Maven Project

✓ Create a simple project

Group Id : com.klu

Artifact Id : springcore-student

Packaging : com.klu



Add Spring Dependency (pom.xml):

```
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>5.3.30</version>
</dependency>
```

Course.java :

```
package com.klu;
public class Course {
    private String courseName;
    private int duration; // duration in months
    // Constructor Injection
    public Course(String courseName, int duration) {
        this.courseName = courseName;
        this.duration = duration;
    }
    // Getters (optional but good practice)
    public String getCourseName() {
        return courseName;
    }
    public int getDuration() {
        return duration;
    }
    @Override
    public String toString() {
        return courseName + " (" + duration + " months)";
    }
}
```

Student.java:

```
package com.klu;
public class Student {
    private int studentId;
    private String name;
    private int year;
    private Course course;
    private String email;
    private String phoneNumber;
    // Constructor Injection
    public Student(int studentId, String name, int year) {
        this.studentId = studentId;
        this.name = name;
        this.year = year;
    }
    // Setter Injection
    public void setCourse(Course course) {
        this.course = course;
    }
}
```

```

    }
    public void setEmail(String email) {
        this.email = email;
    }
    public void setPhoneNumber(String phoneNumber) {
        this.phoneNumber = phoneNumber;
    }
    // Display method (each detail in a separate line)
    public void display() {
        System.out.println("Student ID    : " + studentId);
        System.out.println("Student Name  : " + name);
        System.out.println("Academic Year : " + year);
        System.out.println("Email       : " + email);
        System.out.println("Phone Number : " + phoneNumber);
        System.out.println("Course      : " + course);
    }
}

```

applicationContext.xml:

 src/main/resources/applicationContext.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">

    <!-- Course Bean -->
    <bean id="course" class="com.klu.Course">
        <constructor-arg value="Spring Framework"/>
        <constructor-arg value="6"/>
    </bean>

    <!-- Student Bean -->
    <bean id="student" class="com.klu.Student">

        <!-- Constructor Injection -->
        <constructor-arg value="101"/>
        <constructor-arg value="ram"/>
        <constructor-arg value="2026"/>

        <!-- Setter Injection -->
        <property name="course" ref="course"/>
        <property name="email" value="ram@klu.edu"/>
        <property name="phoneNumber" value="9876543210"/>
    </bean>
</beans>

```

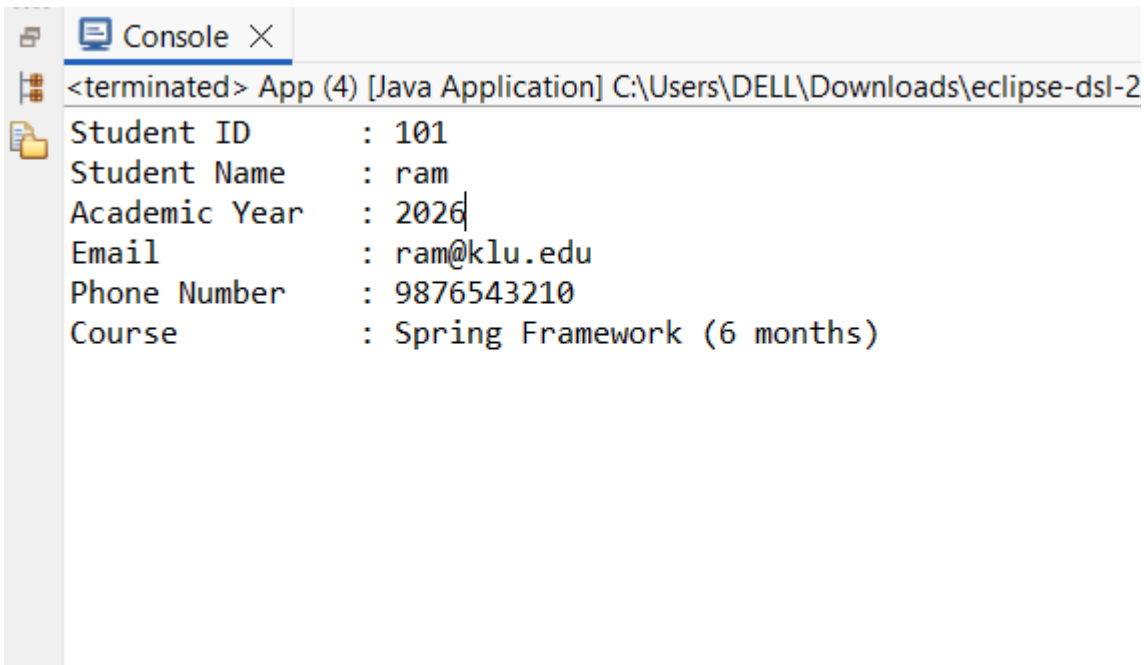
App.java:

```
package com.klu;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
public class App {
    public static void main(String[] args) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");
        Student student = (Student) context.getBean("student");
        student.display();
    }
}
```

Run the Application:

1. Right-click App.java
 2. Run As → Java Application
-

OUTPUT on the console:



```
<terminated> App (4) [Java Application] C:\Users\DELL\Downloads\eclipse-dsl-2
Student ID      : 101
Student Name    : ram
Academic Year   : 2026
Email           : ram@klu.edu
Phone Number    : 9876543210
Course          : Spring Framework (6 months)
```

Now Add the Springcore-students into the Github:

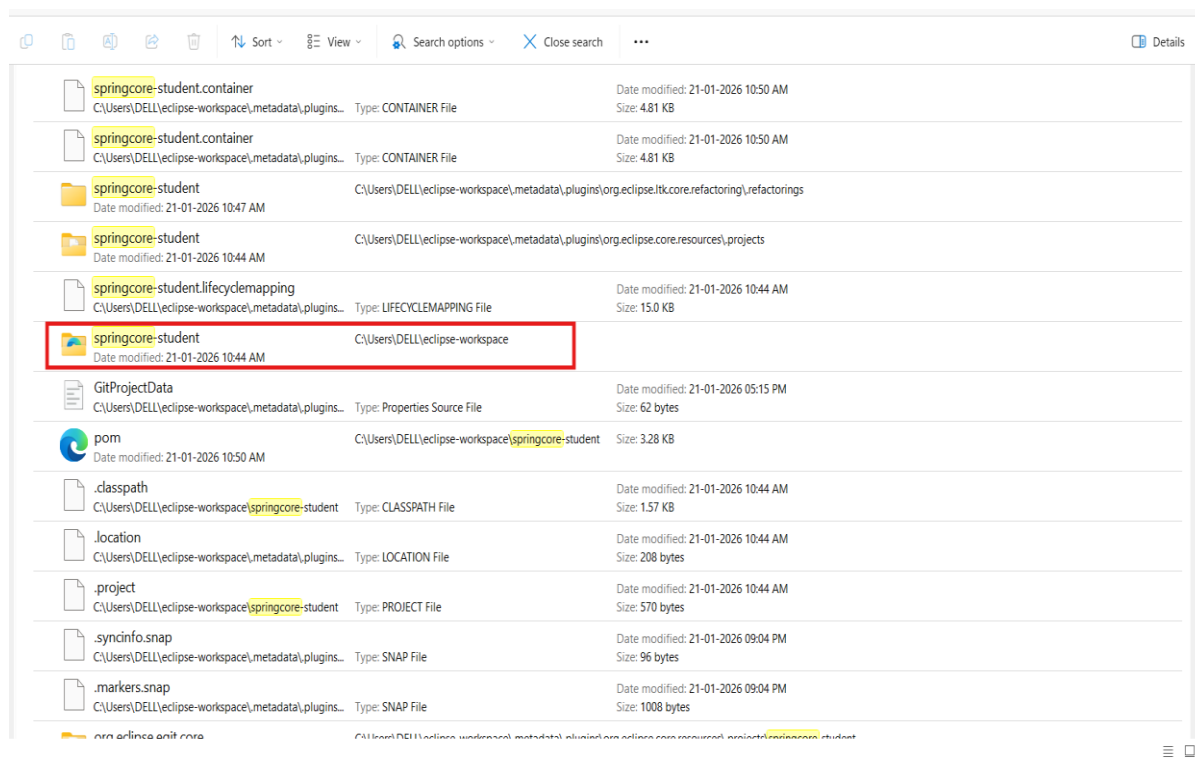
There are 2 ways to push into the Github

1.Through the eclipse

2.Through the files to direct Github.

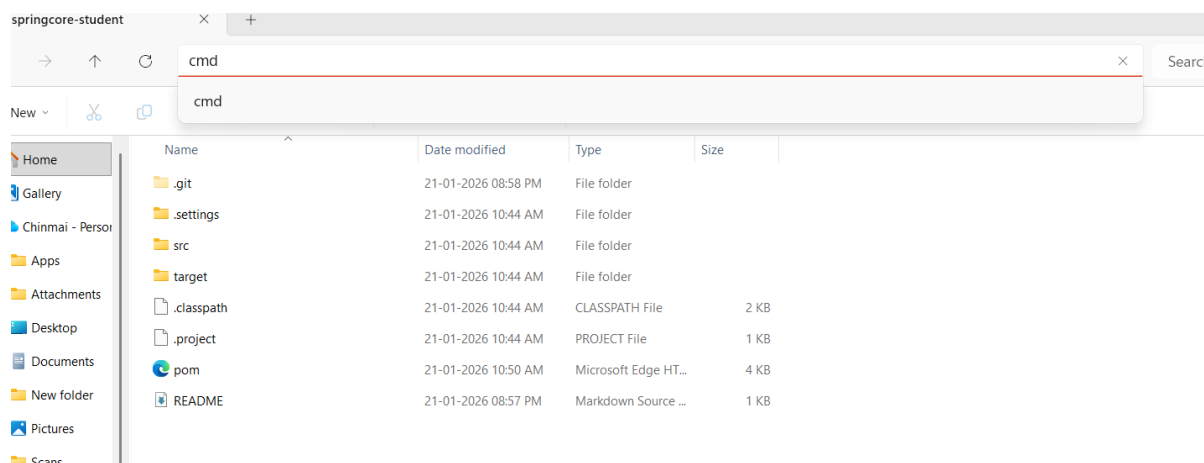
*The easiest method is do it from the files

Now go the file and type springcore-students as shown below



Name	Date modified	Type	Size
.git	21-01-2026 08:58 PM	File folder	
.settings	21-01-2026 10:44 AM	File folder	
src	21-01-2026 10:44 AM	File folder	
target	21-01-2026 10:44 AM	File folder	
.classpath	21-01-2026 10:44 AM	CLASSPATH File	2 KB
.project	21-01-2026 10:44 AM	PROJECT File	1 KB
pom	21-01-2026 10:50 AM	Microsoft Edge HT...	4 KB
README	21-01-2026 08:57 PM	Markdown Source ...	1 KB

1. Keep it aside and Now goto the Github and create a new repository and name it **spring-core-ioc** .
2. After create go the files and type cmd in the search bar



3. Now use the command given the in Github ,so that your files can be uploaded into the Github as shown below or you can upload directly from the Github if cmd didn't work.

```
C:\Windows\System32\cmd.e x + v
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL\eclipse-workspace\springcore-student>
C:\Users\DELL\eclipse-workspace\springcore-student>
C:\Users\DELL\eclipse-workspace\springcore-student>echo "# spring-core-ioc-"
>> README.md

C:\Users\DELL\eclipse-workspace\springcore-student>git init
Initialized empty Git repository in C:/Users/DELL/eclipse-workspace/springco
re-student/.git/

C:\Users\DELL\eclipse-workspace\springcore-student>git add README.md

C:\Users\DELL\eclipse-workspace\springcore-student>git commit -m "first comm
it"
[master (root-commit) dbc3435] first commit
1 file changed, 1 insertion(+)
 create mode 100644 README.md

C:\Users\DELL\eclipse-workspace\springcore-student>git branch -M main

C:\Users\DELL\eclipse-workspace\springcore-student>git remote add origin htt
ps://github.com/RagaChinmai/spring-core-ioc-.git

C:\Users\DELL\eclipse-workspace\springcore-student>git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 239 bytes | 59.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/RagaChinmai/spring-core-ioc-.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\DELL\eclipse-workspace\springcore-student>|
```

